

Jimin Wu

Department of Bioengineering

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RESEARCH INTERESTS

My research integrates **computational imaging, microscopy, neuroengineering, medical imaging** and **machine learning**. By leveraging AI-enabled optics optimization and imaging algorithms, I aim to overcome the challenges inherent in traditional optical systems. Specific projects include developing ultra-compact computational microscopes with superior imaging performance, enabling large-scale neural signal recording and enhanced medical imaging capabilities.

EDUCATION AND EMPLOYMENT

Postdoctoral research associate, Rice University, Houston, TX, USA Present

Electrical and Computer Engineering, George R. Brown School of Engineering

PIs: Jacob T. Robinson, PhD and Ashok Veeraraghavan, PhD

Ph.D. Rice University, Houston, TX, USA July 2025

Bioengineering, George R. Brown School of Engineering

Advisors: Jacob T. Robinson, PhD and Ashok Veeraraghavan, PhD

M.S., Johns Hopkins University, Baltimore, MD, USA June 2019

Electrical and Computer Engineering, Whiting School of Engineering

Advisor: Xingde Li, PhD

B.E., Wuhan University, Wuhan, Hubei, China June 2017

Optical Engineering, School of Electronic Information

PUBLICATIONS

* Equal contribution

- **Jimin Wu***, Huayu Hou*, Jinyun Liu, Vivek Boominathan, Argaja Shende, Karthik Goli, Jennifer Carns, Richard A. Schwarz, Ann M. Gillenwater, Mila P. Salcedo, Kathleen M. Schmeler, Tomasz S. Tkaczyk, Jacob T. Robinson, Ashok Veeraraghavan, Rebecca Richards-Kortum, ‘Dual-modality, deep-learning-enabled endomicroscope with large field-of-view and depth-of-field for real-time in vivo imaging of epithelial precancer’, *Science Advances*, In review (2025)
- Jingyang Zhou, Yuzhi Chen, Matt Whitmire, Pin Kwang Tan, **Jimin Wu**, Ashok Veeraraghavan, Jacob Robinson, Wilson Geisler, Vincent Pieribone, Eyal Seidemann, ‘Fast neural population dynamics in primate V1 captured by a genetically-encoded voltage indicator’, *Proceedings of the National Academy of Sciences*, In review (2025)
- **Jimin Wu***, Yuzhi Chen*, Ashok Veeraraghavan, Eyal Seidemann, Jacob T. Robinson, ‘Mesoscopic calcium imaging in a head-unrestrained male non-human primate using a lensless microscope’, *Nature Communications*, 1-15, 1271 (2024).
- **Jimin Wu**, Vivek Boominathan, Ashok Veeraraghavan, Jacob T. Robinson, ‘Real-time, deep-learning

aided lensless microscope', *Biomedical Optics Express* 8, 4037-4051 (2023).

- **Jimin Wu**, Yuzhi Chen, Ashok Veeraraghavan, Eyal Seidemann, and Jacob T. Robinson "Functional imaging of non-human primate visual cortex using a miniaturized lensless microscope", Proc. SPIE 12365, Neural Imaging and Sensing 2023, 1236504, 2023
- Jesse K. Adams*, Dong Yan*, **Jimin Wu***, Vivek Boominathan*, Sibor Gao, Alex V. Rodriguez, Soonyoung Kim, Jennifer Carns, Rebecca Richards-Kortum, Caleb Kemere, Ashok Veeraraghavan, Jacob T. Robinson, 'In vivo lensless microscopy via a phase mask generating diffraction patterns with high-contrast contours', *Nature Biomedical Engineering*, 1-12, 617-628 (2022)
- **Jimin Wu**, Dong Yan, Vivek Boominathan, Jesse K. Adams, Ashok Veeraraghavan, Jacob T. Robinson, 'Bio-FlatScope: a flat, lensless microscope for fluorescence imaging', Biophotonics Congress 2021, OSA Technical Digest, paper BTh2B. 5, 2021
- Dawei Li*, **Jimin Wu***, Yufan He, Xinwen Yao, Defu Chen, Hyeon-Cheol Park, Kaiyan Li, Wu Yuan, Jerry L. Prince, Xingde Li, 'Parallel deep networks for endoscopic OCT image segmentation', *Biomedical Optics Express* 10, 1126-1135 (2019)
- Jiangfan Liu, **Jimin Wu**, Yun Fang, Xiaoli Xi, 'Factorisation-splitting WLP-FDTD method of wave propagation in dispersive materials', *IET Microwaves, Antennas & Propagation* 15, 1740-1746 (2016)
- Yun Fang, Xiaoli Xi, **Jimin Wu**, Jiangfan Liu, Yurong Pu, 'A JE collocated WLP-FDTD model of wave propagation in isotropic cold plasma', *IEEE Transactions on Microwave Theory and Techniques* 7, 1957-1965 (2016)
- **Jimin Wu**, Xinyue Wang, Yuwei Xie, Jiangfan Liu, 'Ionospheric time-delay of satellite signal propagation calculation based on FDTD method', IEEE Conference on Electromagnetic Field Computation (CEFC), 2016

CONFERENCES & INVITED TALKS

Talks:

- Miniaturized Lensless Microscope for Mesoscopic Calcium Imaging in Head-Unrestrained Non-Human Primates
 - **Invited talk**, 33rd CVS Symposium, Rochester (August 2024)
 - **Invited talk**, 2024 Optica Biophotonics Congress: Biomedical Optics, Fort Lauderdale (April 2024)
- Functional imaging of non-human primate visual cortex using a miniaturized lensless microscope
 - 2023 SPIE Photonics West, San Francisco (January 2023)
 - 2022 Rice Neuroengineering Symposium, Houston (May 2022)
- Bio-FlatScope: a flat, lensless microscope for fluorescence imaging
 - OSA Biophotonics congress: Optics and the Brain, Virtual (April 2021)
- Segmentation of endoscopic OCT images using parallelly trained convolutional neural networks
 - 2019 SPIE Photonics West, San Francisco (February 2019)
- Ionospheric Time-Delay of Satellite Signal Propagation Calculation Based on FDTD Method
 - 17th Biennial IEEE Conference on Electromagnetic Field Computation, Miami (November 2016)

Posters:

- Miniaturized Lensless Microscope for Mesoscopic Calcium and Voltage Imaging in Head-Unrestrained Non-Human Primates
 - Next Generation Computational Bio-Imaging Conference 2024 (**Best Poster Award**)
 - 2024 Society for Neuroscience (SFN) Annual Meeting
- Miniaturized Lensless Microscope for Mesoscopic Calcium Imaging in Head-Unrestrained Non-Human Primates
 - 10th Annual BRAIN Initiative Investigator Meeting
 - 2024 Janelia Computational Optics Conference
 - Harrington Symposium on Optical Methods in Quantitative Bio-imaging: Concept to Application
- In Vivo Calcium Imaging with a Flat, Lensless Microscope
 - 7th Annual BRAIN Initiative Investigator Meeting
- SID - Bio-FlatScope: a Flat, Lensless Imaging Hardware and Software Solution for Calcium Imaging
 - 7th Annual BRAIN Initiative Investigator Meeting

AWARDS

- Rising Star in EECS, MIT 2024
- Rice Future Faculty Fellow 2024
- SPIE Optics and Photonics Education Scholarship 2023
- School of Electronic Information Scholarship, Wuhan University 2014, 2015, 2016

TEACHING EXPERIENCE

Guest Lecture	Intro to Neuroengineering, Rice University	Fall 2023
Teaching Assistant	Thermodynamics, Rice University	Spring 2021
	Biomedical Instrumentation Lab, Rice University	Fall 2020, 2021
	Image Processing & Analysis, Johns Hopkins University	Fall 2018

PROFESSIONAL SERVICE

Reviewer	Nature Publishing: Scientific Reports
	Optica: Optics Express, Applied Optics
	World Scientific Publishing: Journal of Innovative Optical Health Sciences
	PLOS: PLOS ONE

REFERENCES

Prof. Jacob T. Robinson	jtrobinson@rice.edu
Prof. Ashok Veeraraghavan	vashok@rice.edu
Prof. Eyal Seidemann	eyal@austin.utexas.edu